EXPERIMENT-6
SIDDHI SAMBHAV
(24UADS1070)

Objective :- WAP to train and evaluate a Recurrent Neural Network using PyTorch Library to predict the next value in a sample time series dataset

Description :-

Time series data consists of observations collected over time intervals. Unlike traditional models, time series data has temporal dependencies, meaning past values influence future values.

A Recurrent Neural Network (RNN) is a type of neural network designed to handle sequential data. It maintains a hidden state that captures information from previous time steps, making it suitable for time series prediction.

In this experiment:

- **We used a real-world dataset containing features like Revenue and Sales.**
- **The Revenue column was selected as the target variable.**
- **Data was preprocessed using normalization to improve model performance.**
- **The dataset was converted into sequences so that the model can learn patterns over time.**
- **The RNN model was trained to predict the next value based on previous values.**

The model performance was evaluated using:

- **Loss curve (MSE)**
- **Actual vs Predicted graph**
- **Error analysis**

❁ Steps Involved

- **Load the dataset**
- **Clean and preprocess data (handle missing values, normalization)**
- **Create input sequences**
- **Split into training and testing data**
- **Build RNN model using PyTorch**
- **Train the model**
- **Evaluate predictions**
- **Plot graphs for analysis**

Source Code :-

```
import pandas as pd

import numpy as np

import torch

import torch.nn as nn

import matplotlib.pyplot as plt

from sklearn.preprocessing import MinMaxScaler


# 1. Load dataset

df = pd.read_csv("Month_Value_1.csv")


# 2. Clean column names

df.columns = df.columns.str.strip()


# 3. Remove missing values

df = df.dropna()


# 4. Select target column

data = df['Revenue'].astype(float).values.reshape(-1,1)


# 5. Normalize

scaler = MinMaxScaler()
```

```
data = scaler.fit_transform(data)


# 6. Create sequences
def create_sequences(data, seq_len):
    X, y = [], []
    for i in range(len(data) - seq_len):
        X.append(data[i:i+seq_len])
        y.append(data[i+seq_len])
    return np.array(X), np.array(y)


seq_len = 5
X, y = create_sequences(data, seq_len)


# 7. Convert to tensors
X = torch.tensor(X, dtype=torch.float32)
y = torch.tensor(y, dtype=torch.float32)


# 8. Train-test split
train_size = int(0.8 * len(X))
X_train, X_test = X[:train_size], X[train_size:]
y_train, y_test = y[:train_size], y[train_size:]


# 9. Model
class RNNModel(nn.Module):
```

```
    def __init__(self):
        super().__init__()
        self.rnn = nn.RNN(1, 32, batch_first=True)
        self.fc = nn.Linear(32, 1)

    def forward(self, x):
        out, _ = self.rnn(x)
        return self.fc(out[:, -1, :])

model = RNNModel()

# 10. Loss & optimizer
criterion = nn.MSELoss()
optimizer = torch.optim.Adam(model.parameters(), lr=0.001)

# 11. Training
epochs = 10
train_losses = []
val_losses = []

for epoch in range(epochs):
    model.train()

    output = model(X_train)
```

```
    loss = criterion(output, y_train)

    optimizer.zero_grad()
    loss.backward()

    # Prevent exploding gradients
    torch.nn.utils.clip_grad_norm_(model.parameters(), 1)

    optimizer.step()

    train_losses.append(loss.item())

    #  Validation Loss
    model.eval()
    with torch.no_grad():
        val_output = model(X_test)
        val_loss = criterion(val_output, y_test)

    #  PRINT
    print(f"Epoch {epoch+1}, Train Loss: {loss.item()}, Val Loss: {val_loss.item()}")

# 12. Final Predictions
predictions = scaler.inverse_transform(val_output.numpy())
y_test_np = scaler.inverse_transform(y_test.numpy())
```

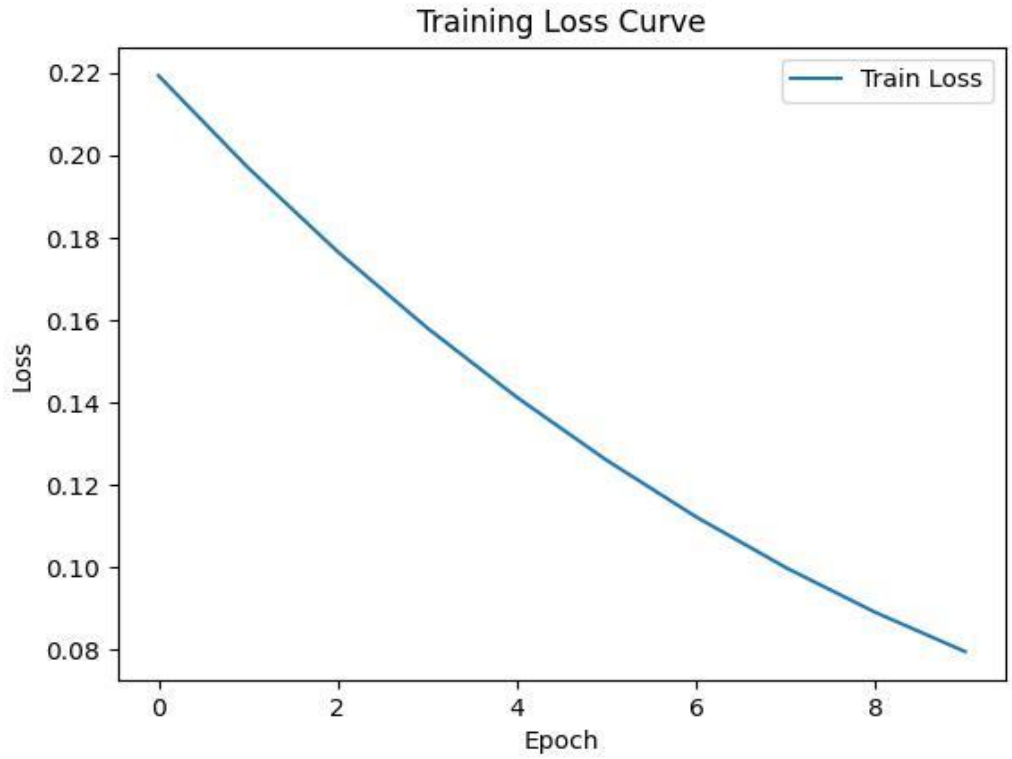
```
# 13. Plot Train vs Validation Loss
plt.figure()
plt.plot(train_losses, label="Train Loss")
plt.legend()
plt.title("Training Loss Curve")
plt.xlabel("Epoch")
plt.ylabel("Loss")
plt.show()

# 14. Plot Predictions
plt.figure()
plt.plot(y_test_np, label="Actual")
plt.plot(predictions, '--', label="Predicted")
plt.legend()
plt.title("Revenue Prediction using RNN")
plt.show()

# 15. Error Plot (bonus)
error = y_test_np - predictions

plt.figure()
plt.plot(error)
plt.title("Prediction Error")
```

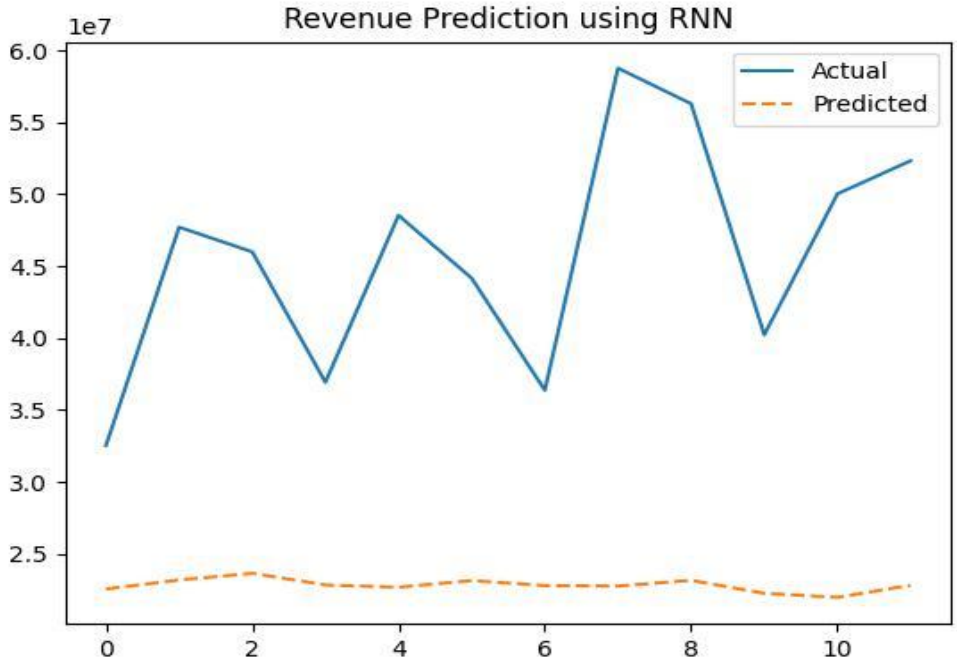
plt.show()

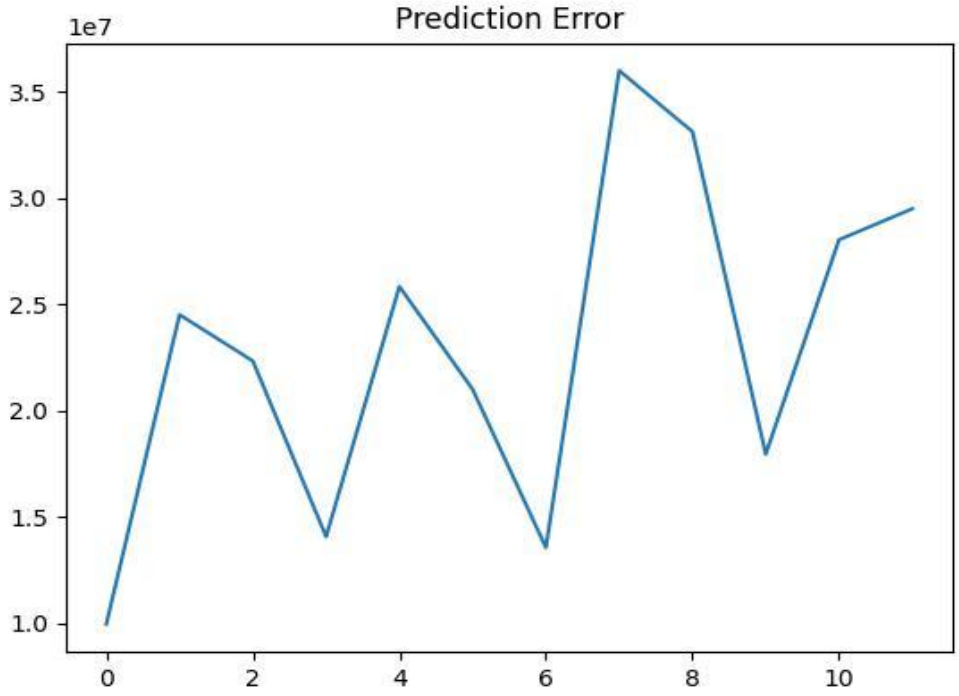
Output:-

```
Epoch 1, Train Loss: 0.37482085824012756, Val Loss: 0.8163308501243591
Epoch 2, Train Loss: 0.34792864322662354, Val Loss: 0.7722181677818298
Epoch 3, Train Loss: 0.32194870710372925, Val Loss: 0.7289478778839111
Epoch 4, Train Loss: 0.2968432903289795, Val Loss: 0.6863986849784851
Epoch 5, Train Loss: 0.2725532650947571, Val Loss: 0.644442617893219
Epoch 6, Train Loss: 0.24903209507465363, Val Loss: 0.6029612421989441
Epoch 7, Train Loss: 0.22625066339969635, Val Loss: 0.5618653893470764
Epoch 8, Train Loss: 0.20420832931995392, Val Loss: 0.5211049914360046
Epoch 9, Train Loss: 0.18293648958206177, Val Loss: 0.48066723346710205
Epoch 10, Train Loss: 0.1624976247549057, Val Loss: 0.44057539105415344
```

GRAPHS : -




Revenue Prediction using RNN
1e7
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
0
2
4
6
8
10
Actual
Predicted


Prediction Error
1e7
3.5
3.0
2.5
2.0
1.5
1.0
0
2
4
6
8
10

MY COMMENTS :-

In this experiment, I learned how Recurrent Neural Networks (RNNs) can be used for time series prediction. At the beginning, I faced some issues like getting NaN values in the loss due to improper data preprocessing and high learning rate. After fixing these problems using normalization and gradient clipping, the model started working properly.

I also understood how important it is to convert the data into sequences, as RNN depends on previous values to predict the next value. By plotting the loss curve and comparing actual and predicted values, I was able to better understand how well the model was performing.

Overall, this experiment helped me gain a better understanding of how deep learning models work with sequential data while using PyTorch for building and training models.